

Jason Chen

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University of Maryland, College Park, MD

RESEARCH INTERESTS

Artificial Intelligence • Machine Learning • Multimodal & Multisensory Intelligence • Computer Vision • Data Science • Software Engineering • Cybersecurity

EDUCATION

University of Maryland, College Park — College Park, MD

Master of Science in Computer Science | August 2026 – Present

University of Maryland, College Park — College Park, MD

Bachelor of Science in Computer Science | August 2023 – May 2026

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, SQL, HTML

AI & Machine Learning: Minimax, Negamax, Alpha-Beta Pruning, Constraint-Based Search, AI Optimization

Tools & Platforms: AWS, Git, Visual Studio Code, Wireshark, Virtual Machines, Backstage.io, Kubernetes, Ansible

Operating Systems: Windows, macOS, Linux

Research Areas: Artificial Intelligence, Machine Learning, Multisensory Intelligence, Data Science, Cybersecurity, Software Engineering, Databases

RESEARCH EXPERIENCE

Zeta Associates — Research/Software Engineering Intern

Fairfax, VA | May 2025 – August 2025 | On-site

- Led the research, implementation, and deployment of an internal developer platform (IDP) using **Backstage.io**, investigating approaches for improving developer workflows, service discoverability, and self-service capabilities.
- Designed and implemented integrations between the developer platform and existing infrastructure automation, enabling automated **Kubernetes cluster provisioning** and software deployment on virtual desktops.
- Connected existing **Ansible playbooks** to the developer platform to automate critical infrastructure tasks and reduce manual operational processes.

- Evaluated and refined platform functionality to improve developer productivity, software delivery efficiency, and infrastructure accessibility.
- Worked across software development, infrastructure automation, and cloud-native technologies in a collaborative engineering environment.

Thomas Jefferson High School for Science and Technology — AI Research Projects

Alexandria, VA

Othello AI

- Developed a two-player **Othello board game in Python** with an AI opponent based on **Minimax/Negamax search and Alpha-Beta pruning**.
- Optimized game-tree computation to improve AI decision-making performance and reduce unnecessary search operations.
- Implemented an **opening book** based on Othello game theory to improve decision quality during the initial stage of gameplay.
- Explored algorithmic tradeoffs between search depth, computational efficiency, and playing strength.

Sudoku Solver

- Developed a Python-based Sudoku solver capable of solving large numbers of puzzles through automated search and constraint-based reasoning.
- Improved upon brute-force search by restricting candidate values using **Sudoku constraints and logical pruning techniques**.
- Further optimized the solver through algorithmic improvements, enabling it to solve approximately **2,000 puzzles within five minutes**.
- Investigated the impact of constraint propagation and search-space reduction on computational performance.

WORK EXPERIENCE

Software Engineer Intern — Roblox

June 2026 – Present

- Contribute to software engineering projects involving application development, testing, debugging, and implementation.
- Apply programming and problem-solving skills to develop and improve software systems.

- Collaborate with team members to troubleshoot technical issues and support software development activities.

AI Intern — Criterion Systems

Tysons Corner, VA | June 2024 – August 2024 | Hybrid

- Developed and maintained **AI-powered chatbot systems** designed to provide efficient and accurate customer support.
- Monitored chatbot performance and proactively identified and resolved system issues through testing and iterative improvements.
- Documented intelligent automation implementations to support knowledge transfer, maintainability, and future system scalability.
- Provided operational and technical support for AI-enabled business processes and initiatives.
- Collaborated with team members to evaluate system behavior and improve the reliability and effectiveness of AI-driven solutions.

ACADEMIC HONORS & AWARDS

- **President's Scholarship** — Four-year recipient
- **Multiple Academic Achievement Awards**
- **Dean's List**
- **Honors Program Admittance**

PROFESSIONAL & RESEARCH GOALS

Pursue graduate-level research in **artificial intelligence, machine learning, and multisensory/multimodal machine intelligence**, with an emphasis on developing intelligent systems that can learn from and integrate information from multiple sources and modalities. Interested in combining strong algorithmic foundations with practical system development to investigate challenging problems in AI and contribute to impactful research through faculty collaboration and interdisciplinary work.